03.09.2021

3 Eylül 2021 Cuma 08:03

Q3) (35 p.)For the network in figure below, determine the following:

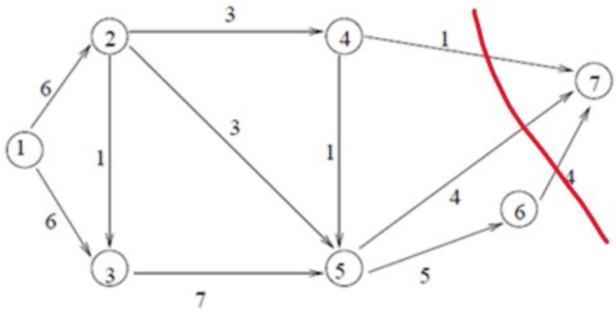
a) The maximum flow from node 1 to node 7 by applying the necessary algorithm.

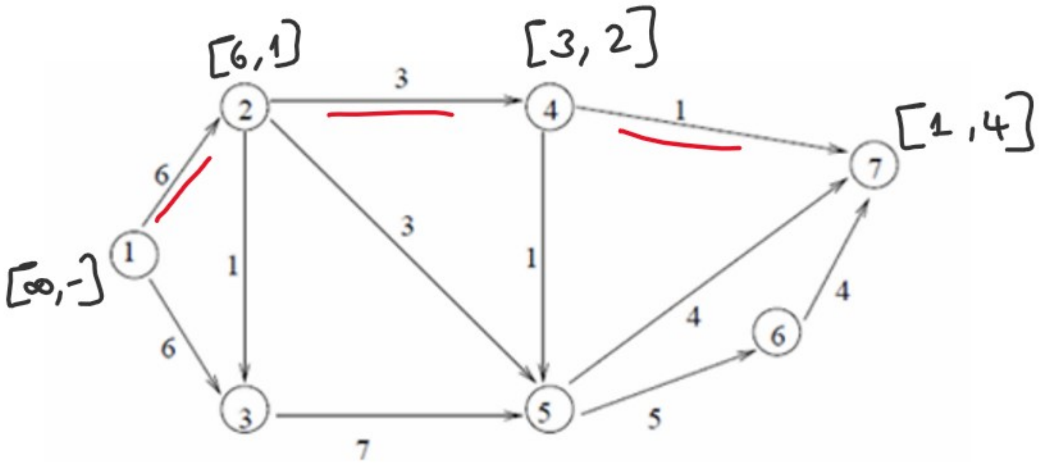
b) The surplus capacities for all arcs.

c) The amount of flow through the nodes 2, 3, 4, 5 and 6.

d) A cut whose capacity equals the maximum flow in the network.

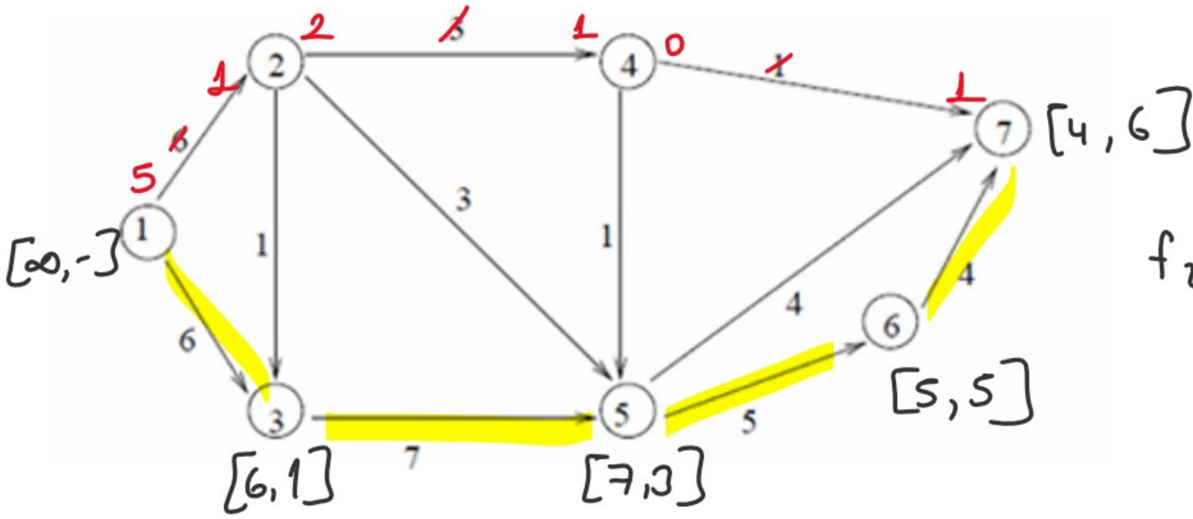
$S = \{1,2,3,4,5,6\}$ $\bar{S} = \{ 7. \}$





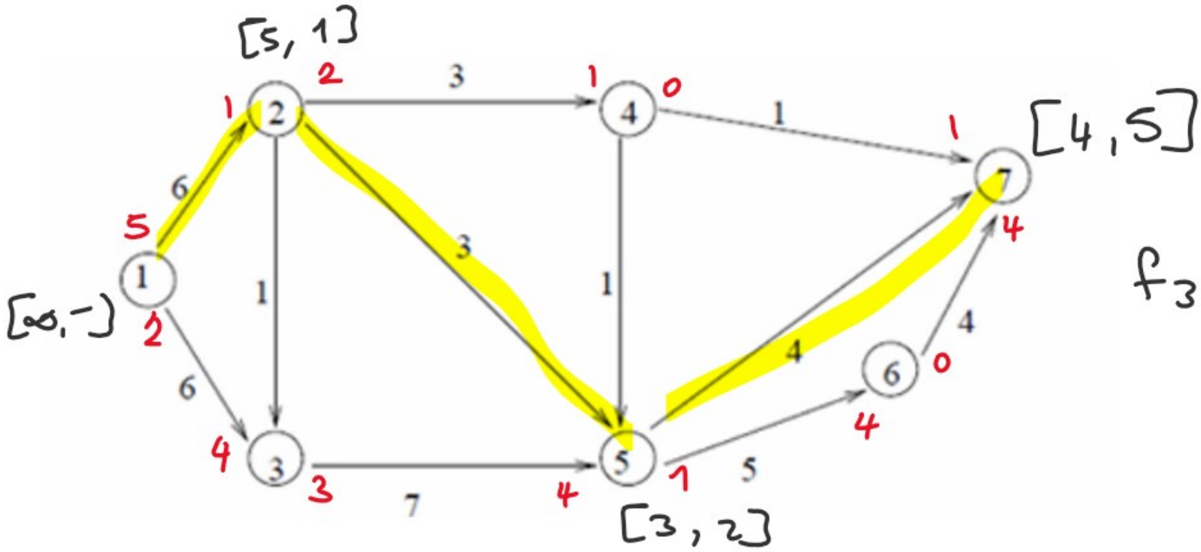
$f_1 = \min\{\infty, 6, 3, 1\}$

$f_1 = 1.$

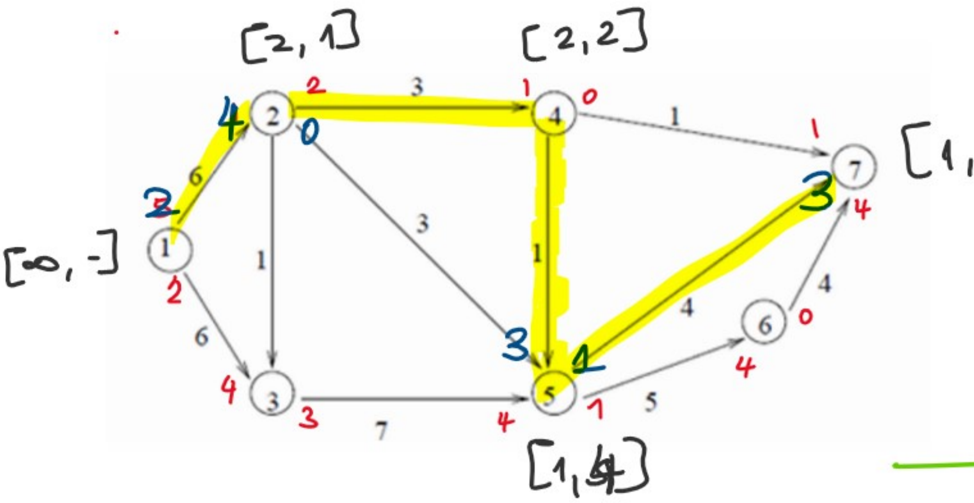


$f_2 = 4.$

[5, 1]

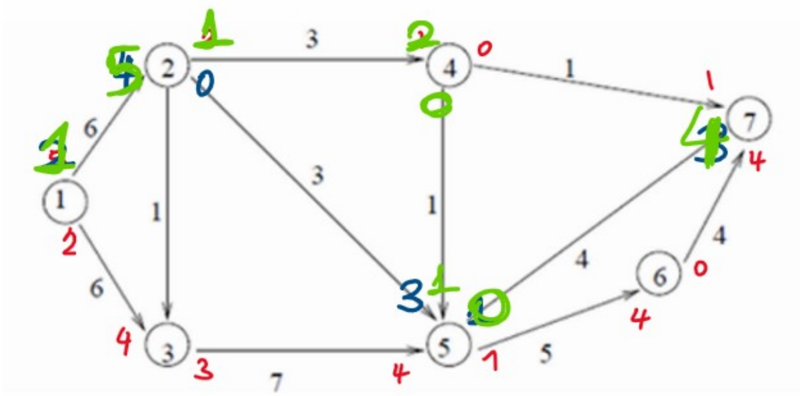


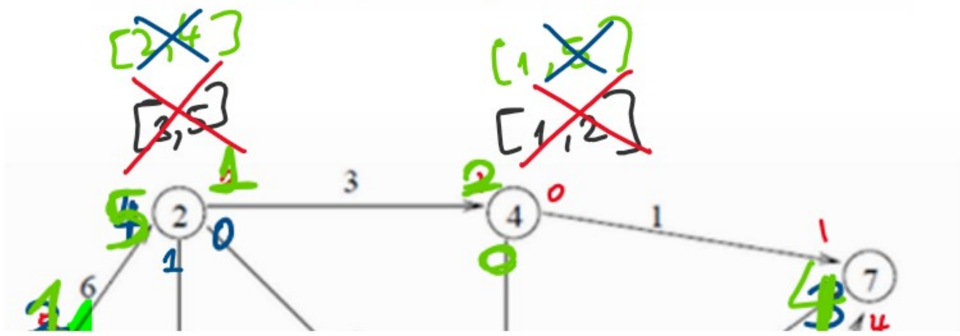
$f_3 = 3.$

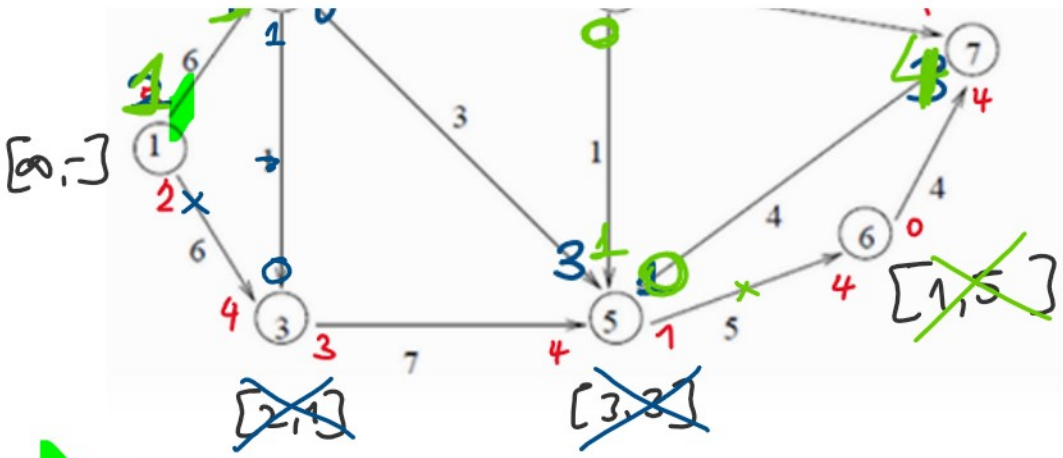


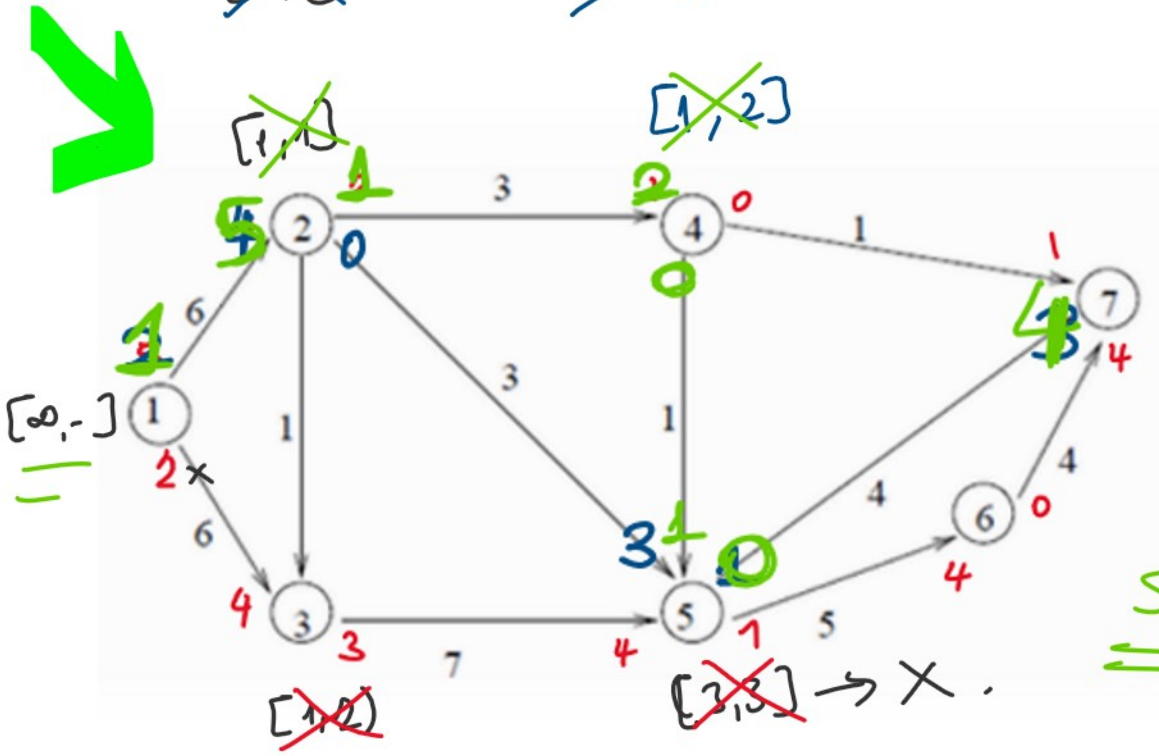
$f_4 = 1.$

$\boxed{F = 9}$









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